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1 考点 1: 波动率微笑

1. 一个欧式看涨期权的市场价格为 \$3.00，其 Black-Scholes 模型价格为 \$3.50。具有相同执行价格和到期时间的欧式看跌期权的 Black-Scholes 模型价格为 \$2.00。该看跌期权的市场价格应该是多少？

- A. \$1.50
- B. \$2.00
- C. \$1.00
- D. \$0.50

答案:A

解析:

A 选项正确。

- 根据 看涨-看跌平价关系（Put-Call Parity），我们有：

$$c_{BS} + Ke^{-rT} = p_{BS} + S_0e^{-qT}$$

$$c_{mkt} + Ke^{-rT} = p_{mkt} + S_0e^{-qT}$$

两式相减可得：

$$c_{BS} - c_{mkt} = p_{BS} - p_{mkt}$$

已知：

$$c_{BS} = 3.50, \quad c_{mkt} = 3.00, \quad p_{BS} = 2.00$$

代入上式：

$$3.50 - 3.00 = 2.00 - p_{mkt} \Rightarrow p_{mkt} = 1.50$$

因此，该看跌期权的市场价格应为：

1.50

B 选项错误。

- 这是 Black-Scholes 模型价格，而不是市场价格。

C 选项错误。

- 计算错误，期权价值被低估。

D 选项错误。

- 计算错误，期权价值被低估。

考察: Black-Scholes 模型中的看涨-看跌期权平价关系

2. The ’ Sticky Strike Rule’ assumes that implied volatility is:

- A. The same across maturities for given strike prices
- B. The same for short time periods
- C. The same across strike prices for given maturities
- D. Different across strike prices for given maturities

答案:B

解析:

A 选项错误。

- Sticky Strike Rule 并不假设隐含波动率在不同到期日之间保持不变。不同到期日的期权通常会有不同的隐含波动率。

B 选项正确。

- Sticky Strike Rule 的核心假设是：在计算期权敏感性指标时，隐含波动率在短期内保持不变。这个假设使得我们可以更准确地计算期权的风险指标。

C 选项错误。

- Sticky Strike Rule 并不假设隐含波动率在不同行权价格之间保持不变。实际上，不同行权价格的期权通常会有不同的隐含波动率。

D 选项错误。

- 虽然这个选项描述了波动率微笑现象，但这不是 Sticky Strike Rule 的核心假设。该规则主要关注时间维度上的波动率稳定性。

考察: Sticky Strike Rule 假设隐含波动率在短期内保持不变

3. John, FRM, uses the Black-Scholes-Merton (BSM) model to value options. Following the financial crisis of 2008-2010, he is more aware of the limitations of the BSM option pricing model. Which of the following statements best characterizes a major limitation of the BSM option pricing model?

- A. The BSM model assumes strike prices have non-constant volatility.
- B. Option traders often use a volatility smile with lower volatilities for out-of-the-money call and put options when applying the BSM model.
- C. For up-and-out calls and puts, the BSM model is insensitive to changes in implied volatility when the knock-out strike price is equal to the strike price and the interest rate equals the underlying asset return.
- D. For down-and-out calls and puts, the BSM model is insensitive to changes in option maturity when the knock-out strike price is greater than the strike price and the interest rate is greater than the underlying asset return.

答案:C

解析:

A 选项错误。

- BSM 模型实际上假设波动率是恒定的，而不是非恒定的。这是模型的一个基本假设。

B 选项错误。

- 期权交易员通常对价外期权使用更高的波动率，而不是更低的波动率，这形成了波动率微笑现象。

C 选项正确。

- 对于上升出局期权，当敲出价格等于行权价格且利率等于标的资产收益率时，BSM 模型对隐含波动率的变化不敏感。这是模型的一个重要局限性。

D 选项错误。

- 对于下降出局期权，BSM 模型对期权期限的变化是敏感的，特别是在敲出价格大于行权价格且利率大于标的资产收益率的情况下。

考察: BSM 模型对上升出局期权的隐含波动率变化不敏感

4. Which of the following statements is incorrect regarding volatility smiles?

- A. Currency options exhibit volatility smiles because the at-the-money options have higher implied volatility than away-from-the-money options.
- B. Volatility frowns result when jumps occur in asset prices.
- C. Equity options exhibit a volatility smirk because low strike price options have greater implied volatility.
- D. Relative to currency traders, it appears that equity traders' expectations of extreme price movements are more asymmetric.

答案:A

解析:

A 选项正确。

- 货币期权表现出波动率微笑，是因为平价期权的隐含波动率低于价外期权，而不是高于价外期权。这是货币期权市场的典型特征。

B 选项错误。

- 当资产价格出现跳跃时，确实会导致波动率 frown（倒微笑）现象。这是正确的描述。

C 选项错误。

- 股票期权确实表现出波动率 smirk（偏斜），其中低行权价格的期权具有更高的隐含波动率。这反映了市场对大幅下跌的担忧。

D 选项错误。

- 相对于货币交易者，股票交易者对极端价格变动的预期确实更加不对称。这反映了两个市场的不同特征。

考察: 货币期权平价期权的隐含波动率低于价外期权

5. Compared to at-the-money currency options, out-of-the-money currency options exhibit which of the following volatility traits?

- A. Lower implied volatility
- B. A frown
- C. A smirk
- D. Higher implied volatility

答案:D

解析:

A 选项错误。

- 价外货币期权的隐含波动率实际上高于平价期权，而不是低于平价期权。这是货币期权市场的典型特征。

B 选项错误。

- 波动率 frown（倒微笑）通常出现在资产价格发生跳跃时，而不是货币期权的典型特征。

C 选项错误。

- 波动率 smirk（偏斜）是股票期权的典型特征，而不是货币期权的特征。货币期权通常表现出对称的波动率微笑。

D 选项正确。

- 与平价期权相比，价外货币期权具有更高的隐含波动率。这种模式形成了波动率微笑，反映了市场对大幅价格变动的预期。

考察: 货币期权价外期权的隐含波动率高于平价期权

6. Which of the following regarding equity option volatility is true?

- A. There is higher implied price volatility for away-from-the-money equity options
- B. ” Crashophobia” suggests actual equity volatility increases when stock prices decline
- C. Compared to the lognormal distribution, traders believe the probability of large down movements in price is similar to large up movements
- D. Increasing leverage at lower equity prices suggests increasing volatility

答案:D

解析:

A 选项错误。

- 股票期权的波动率特征是对称的，价外期权的隐含波动率并不一定高于平价期权。这与货币期权市场不同。

B 选项错误。

- ” Crashophobia”（崩盘恐惧）是指市场认为大幅下跌的可能性高于 BSM 模型假设，而不是指价格下跌时实际波动率会增加。

C 选项错误。

- 与对数正态分布相比，交易员认为价格大幅下跌的概率高于大幅上涨的概率，而不是相似。这反映了市场对下跌风险的担忧。

D 选项正确。

- 当股票价格较低时，公司杠杆率增加，这会导致波动率增加。这是股票期权市场的一个重要特征。

考察: 股票价格下跌时杠杆增加导致波动率上升

7. The chief risk officer of Alpha Investments Group is planning a change in methodology for some of the risk management models used to estimate risk measures. His aim is to move from models that use the normal distribution of returns to models that use the distribution of returns implied by market prices. Alpha Group has a large long position in the Japanese equity stock index Nikkei, which has a volatility smile that slopes downward to the right. How will the change in methodology affect the estimate of expected shortfall?

- A. ES with the updated models will be larger than the old estimate
- B. ES with the updated models will be smaller than the older estimate
- C. ES will remain unchanged
- D. Insufficient information to determine

答案:A

解析:

A 选项正确。

- 由于波动率微笑向右下方倾斜，这表明隐含分布比对数正态分布有更厚的左尾。这意味着在隐含分布下，日经指数发生极端下跌的概率更高。因此，使用新方法计算的 ES 值会更大。

B 选项错误。

- 更厚的左尾意味着更大的下行风险，因此 ES 值不会变小。

C 选项错误。

- 从正态分布转换到隐含分布会改变风险估计，ES 值不会保持不变。

D 选项错误。

- 题目提供了足够的信息：波动率微笑向右下方倾斜，表明隐含分布有更厚的左尾，这足以判断 ES 值的变化方向。

考察: 波动率微笑右下方倾斜时隐含分布左尾更厚，ES 值更大

8. A risk manager is in the process of valuing several European option positions on a non-dividend-paying stock ABC that is currently priced at USD 50. The implied volatility skew, estimated using the Black-Scholes-Merton model and the current prices of actively traded European-style options on stock ABC at various strike prices, is:

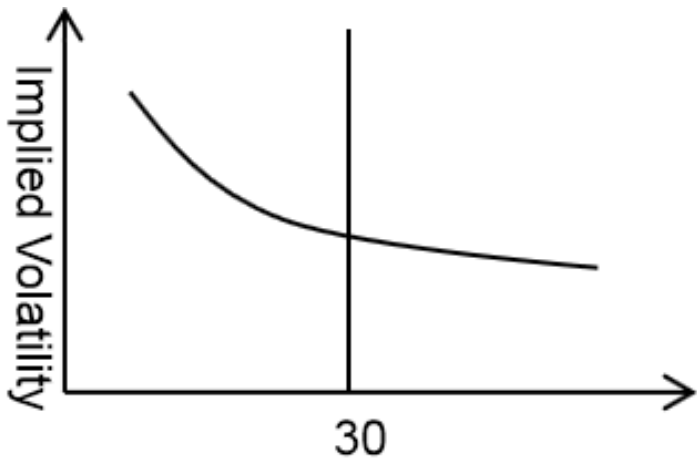


Figure 1: 示例图片

Assuming that the implied volatility at USD 50 is used to conduct the valuation, which of the following long positions will be undervalued?

- A. An out-of-the-money call
- B. An in-the-money call
- C. An at-the-money put

D. An in-the-money put

答案:B

解析:

A 选项错误。

- 价外看涨期权的行权价格高于 50，其隐含波动率低于平价波动率，使用平价波动率会导致高估而不是低估。

B 选项正确。

- 价内看涨期权的行权价格低于 50，其隐含波动率高于平价波动率。使用平价波动率会导致期权价值被低估。

C 选项错误。

- 平价看跌期权的行权价格等于 50，使用平价波动率进行估值是准确的，不会产生低估。

D 选项错误。

- 价内看跌期权的行权价格高于 50，其隐含波动率低于平价波动率，使用平价波动率会导致高估而不是低估。

考察: 使用平价波动率会导致价内看涨期权被低估

9. With all other things being equal, a risk monitoring system that assumes constant volatility for equity returns will understate the implied volatility for which of the following positions by the largest amount:

- A. Short position in an at-the-money call
- B. Long position in an at-the-money call
- C. Short position in a deep out-of-the-money call
- D. Long position in a deep in-the-money call

答案:D

解析:

A 选项错误。

- 平价看涨期权的隐含波动率与假设的恒定波动率差异较小，因此低估程度也较小。

B 选项错误。

- 平价看涨期权的隐含波动率与假设的恒定波动率差异较小，因此低估程度也较小。

C 选项错误。

- 深度价外看涨期权的隐含波动率通常低于平价期权，使用恒定波动率会导致高估而不是低估。

D 选项正确。

- 深度价内看涨期权的隐含波动率显著高于平价期权。当风险监控系統假设恒定波动率时，会最大程度地低估这类期权的隐含波动率。

考察: 假设恒定波动率会最大程度低估深度价内看涨期权的隐含波动率

10. Michael Chen, FRM, is a hedge fund manager who is explaining implied volatility for currency options to junior analysts. Which of the following statements best completes his explanation?

- A. There is a greater chance of extreme price movements than predicted by a lognormal distribution
- B. Arbitrage opportunities clearly exist
- C. The implied volatility of currency options is expected to increase in the near future
- D. There are no arbitrage opportunities unless the implied volatility versus strike price represents a skewness that is referred to as a smirk rather than a smile

答案:A

解析:

A 选项正确。

- 如果货币期权的价外期权隐含波动率高于平价期权，这表明交易者认为极端价格变动的概率大于对数正态分布预测的概率。实证研究支持这一假设。

B 选项错误。

- 波动率微笑现象本身并不表明存在套利机会。市场参与者对极端价格变动的预期差异是导致波动率微笑的原因。

C 选项错误。

- 波动率微笑是描述隐含波动率与行权价格关系的静态特征，而不是对未来波动率变化的预测。

D 选项错误。

- 波动率微笑和波动率偏斜（smirk）都是市场对极端价格变动预期的反映，与套利机会的存在与否无关。

考察: 货币期权的波动率微笑表明极端价格变动的概率大于对数正态分布预测

11. David Wang, FRM, uses the Black-Scholes-Merton (BSM) model to value options. Following the financial crisis of 2008-2010, he is more aware of the limitations of the BSM option pricing model. Which of the following statements best characterizes a major limitation of the BSM option pricing model?

- A. The BSM model assumes strike prices have non-constant volatility
- B. Option traders often use a volatility smile with lower volatilities for out-of-the-money call and put options when applying the BSM model
- C. For up-and-out calls and puts, the BSM model is insensitive to changes in implied volatility when the knock-out strike price is equal to the strike price and the interest rate equals the underlying asset return
- D. For down-and-out calls and puts, the BSM model is insensitive to changes in option maturity when the knock-out strike price is greater than the strike price and the interest rate is greater than the underlying asset return

答案:C

解析:

A 选项错误。

- BSM 模型实际上假设波动率是恒定的，而不是非恒定的。这是模型的一个基本假设。

B 选项错误。

- 期权交易员通常对价外期权使用更高的波动率，而不是更低的波动率，这形成了波动率微笑现象。

C 选项正确。

- 对于上升出局期权，当敲出价格等于行权价格且利率等于标的资产收益率时，BSM 模型对隐含波动率的变化不敏感。这是模型的一个重要局限性。

D 选项错误。

- 对于下降出局期权，BSM 模型对期权期限的变化是敏感的，特别是在敲出价格大于行权价格且利率大于标的资产收益率的情况下。

考察: BSM 模型对上升出局期权的隐含波动率变化不敏感

12. An European call with a market price of \$3.5 has a BSM price of \$4.5. All else equal, the BSM price of a European put is \$2.5. What is the market price of the put?

- A. \$2
- B. \$2.5
- C. \$1.5
- D. \$1

答案:C

解析:

A 选项错误。

- 根据看涨-看跌期权平价关系，市场价格的差异应该等于 BSM 价格的差异。 $3.5 - 4.5 = -1$ ，因此看跌期权的市场价格应该是 $2.5 - 1 = 1.5$ ，而不是 2。

B 选项错误。

- 2.5 是 BSM 模型计算出的看跌期权价格，而不是市场价格。市场价格需要考虑 BSM 模型与市场价格的差异。

C 选项正确。

- 根据看涨-看跌期权平价关系： $C_{mkt} - C_{BSM} = P_{mkt} - P_{BSM}$ ，代入数值： $3.5 - 4.5 = P_{mkt} - 2.5$ ，解得 $P_{mkt} = 1.5$ 。

D 选项错误。

- 1.5 是正确答案，1 明显小于正确值。

考察: 看涨-看跌期权平价关系在 BSM 模型中的应用

13. Lisa, the Risk Market Analyst, calculated implied volatilities based on call options for Apple’ s traded equity. As it turns out, the shape of the corresponding implied volatility skew is somewhat typical of equity options; i.e., downward sloping per decreasing implied volatility as a function of increasing strike price. She draws the following conclusions:

1. The implied distribution of Apple’ s stock price has negative skew and negative excess kurtosis
2. For identical maturities, an in-the-money call (ITM call) option on Apple’ s stock is more expensive than an out-of-the-money put (OTM put) option
3. Compared to the normal distribution, the implied distribution of Apple’ s stock price has a lighter left tail
4. Compared to the lognormal distribution, the implied distribution of Apple’ s stock price has a lighter right tail

Which of her above conclusions is necessarily true?

- A. None
- B. I only
- C. IV only
- D. II and III only

答案:C

解析:

A 选项错误。

- 结论 IV 是正确的，因此” None” 不正确。

B 选项错误。

- 结论 I 错误。股票期权的波动率偏斜表明隐含分布有负偏度，但通常有正超额峰度，而不是负超额峰度。

C 选项正确。

- 结论 IV 正确。与对数正态分布相比，隐含分布的右尾更轻，这反映了市场对大幅上涨的预期较低。

D 选项错误。

- 结论 II 和 III 都是错误的。价内看涨期权不一定比价外看跌期权更贵，且隐含分布的左尾通常比正态分布更重而不是更轻。

考察: 股票期权的隐含分布右尾比对数正态分布更轻

14. A committee of risk management practitioner discusses the difference between pricing deep out-of-the-money call options on TSLA stock and pricing deep out-of-the-money call options on the USD/CNY foreign exchange rate using the Black-Scholes-Merton (BSM) model. The practitioners price these options based on two distinct probability distributions of underlying asset prices at the option expiration date:

- A lognormal probability distribution
- An implied risk-neutral probability distribution obtained from the volatility smile for options of the same maturity

Using the lognormal instead of the implied risk-neutral probability distribution will tend to:

- A. Price the option on TSLA relatively high and price the option on USD/CNY relatively low
- B. Price the option on TSLA relatively low and price the option on USD/CNY relatively high
- C. Price the option on TSLA relatively low and price the option on USD/CNY relatively low
- D. Price the option on TSLA relatively high and price the option on USD/CNY relatively high

答案:A

解析:

A 选项正确。

- 股票期权的隐含分布比对数正态分布有更重的左尾和更轻的右尾，因此使用对数正态分布会导致深度价外看涨期权被高估。而货币期权的隐含分布比对数正态分布有更重的尾部，因此使用对数正态分布会导致深度价外看涨期权被低估。

B 选项错误。

- 这个选项完全颠倒了股票和货币期权的定价偏差方向。

C 选项错误。

- 这个选项错误地认为两种期权都会被低估。

D 选项错误。

- 这个选项错误地认为两种期权都会被高估。

考察: 对数正态分布会高估股票期权而低估货币期权

15. You are asked to mark to market a book of plain-vanilla stock options. The trader is short deep out-of-the-money options and long at-the-money options. There is a pronounced smile for these options. The trader’s bonus increases as the value of his book increases. Which approach should you use to mark the book?

- A. Use the implied volatility of at-the-money options because the estimation of the volatility is more reliable
- B. Use the average of the implied volatilities for the traded options for which you have data because it provides a balanced view
- C. For each option, use the implied volatility of the most similar option traded on the market
- D. Use the historical volatility because doing so corrects for the pricing mistakes in the option mark

答案:C

解析:

A 选项错误。

- 使用平价期权的隐含波动率会低估价外期权的价值，这会人为地增加交易员的利润，不符合公允定价原则。

B 选项错误。

- 使用平均隐含波动率会忽略波动率微笑的特征，无法准确反映不同行权价格期权的实际市场价值。

C 选项正确。

- 对每个期权使用市场上最相似期权的隐含波动率，可以最准确地反映其市场价值。这种方法考虑了波动率微笑的特征，能够给出最接近市场价格的估值。

D 选项错误。

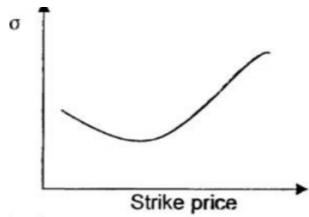
- 历史波动率无法反映当前市场对未来价格变动的预期，不适合用于期权定价。

考察: 使用最相似期权的隐含波动率进行期权定价

16. Suppose that a central bank’s policy is to allow an exchange rate to fluctuate between 0.95 and 1.05. What pattern of implied volatilities for options on the exchange rate would you expect to see?



A.



B.



C.



D.

答案:A

解析:

A 选项正确。

- 由于汇率波动范围被限制在 0.95 到 1.05 之间，这意味着汇率的最小值为 0.95，最大值为 1.05。与对数正态分布相比，汇率的隐含概率分布有更薄的左尾和右尾。因此，价外和价内期权的隐含波动率低于平价期权。

B 选项错误。

- 这个选项显示的是波动率微笑，表示价外期权的隐含波动率高于平价期权，这与汇率波动范围限制的情况不符。

C 选项错误。

- 这个选项显示的是波动率偏斜，表示低行权价格期权的隐含波动率高于高行权价格期权，这与汇率波动范围限制的情况不符。

D 选项错误。

- 这个选项显示的是波动率 frown（倒微笑），表示平价期权的隐含波动率高于价外期权，这与汇率波动范围限制的情况不符。

考察: 汇率波动范围限制导致价外和价内期权的隐含波动率低于平价期权

17. An empirical distribution of equity price derived from the price of options of such stock based on BSM that exhibits a fatter right tail than that of a lognormal distribution would indicate:

A. Equal implied volatilities across low and high strike prices

- B. Greater implied volatilities for low strike prices
- C. Greater implied volatilities for high strike prices
- D. Higher implied volatilities for mid-range strike prices

答案:C

解析:

A 选项错误。

- 如果隐含波动率在低行权价格和高行权价格之间相等，这表明隐含分布与对数正态分布相似，没有表现出厚尾特征。

B 选项错误。

- 低行权价格期权的隐含波动率更高通常表明隐含分布有厚左尾，而不是厚右尾。

C 选项正确。

- 具有厚右尾的隐含分布表明观察到高标的资产价格的概率增加，这会导致高行权价格期权的隐含波动率更高。

D 选项错误。

- 中等行权价格期权的隐含波动率更高通常表明隐含分布有更厚的中间部分，而不是厚右尾。

考察: 厚右尾隐含分布导致高行权价期权隐含波动率更高

18. A newly hired junior derivatives pricing analyst at an investment bank is evaluating option contracts on the USD/JPY exchange rate. The analyst observes that the risk-free interest rate is 3% in the US and 0.1% in Japan and gathers the following information on four European-style at-the-money options:

Option	Option type	Days to expiration	Option price
1	Call	30	2.15
2	Put	30	1.82
3	Call	90	3.85
4	Put	90	3.65

The analyst uses the Black-Scholes-Merton model to calculate the implied volatilities of the options. Assuming no arbitrage opportunities, which of the following would the analyst be correct to conclude about these options?

- A. The implied volatility of the options expiring in 90 days is higher than that of the options expiring in 30 days due to the interest rate differential between the two countries
- B. Put-call parity does not hold for these currency options since the prices of the at-the-money call options are not equal to the prices of the at-the-money put options
- C. The implied volatility of the at-the-money put option expiring in 30 days is the same as the implied volatility of the at-the-money call option expiring in 30 days
- D. The volatility smiles for options on the USD/JPY exchange rate display positive upward skewness since the at-the-money option prices clearly display a preference for calls over puts

答案:C

解析:

A 选项错误。

- 利率差异会影响期权价格，但不会直接影响隐含波动率。隐含波动率主要反映市场对未来价格波动的预期。

B 选项错误。

- 看涨-看跌期权平价关系（Put-call parity）仍然成立。平价期权中看涨和看跌期权的价格不同是由于利率差异导致的，这并不违反平价关系。

C 选项正确。

- 对于具有相同行权价格和到期日的欧式期权，看涨期权的隐含波动率总是等于看跌期权的隐含波动率。这是因为它们都反映了同一标的资产的价格波动预期。

D 选项错误。

- 平价期权中看涨期权价格高于看跌期权价格是由于利率差异导致的，而不是波动率微笑的偏斜特征。

考察: 相同行权价和到期日的欧式期权隐含波动率相等

19. A pharmaceutical company plans to announce the results of a clinical trial in three months, which is expected to cause the price of its stock to move sharply, either up or down. A hedge fund manager would like to take advantage of this event by establishing an options position. In preparation, the manager examines the volatility curves, volatility term structure, and volatility surface of the options on the stock. Which of the following is correct for the manager to find through this examination?

- A. The volatility surface will reflect implied volatilities being an increasing function of time to maturity, regardless of the shape of the volatility curves
- B. The volatility curve at each maturity will steepen as the time to maturity of the options increases, regardless of the shape of the volatility curves
- C. The volatility curves of short-dated options that mature after the event form a steep smile, while the volatility curves of options with longer-dated maturities are flat
- D. The volatility curves for options with maturities of three months or longer are frown-shaped and the term structure of implied volatility is a decreasing function of time to maturity

答案:C

解析:

A 选项错误。

- 波动率曲面不一定总是到期时间的增函数。在重大事件（如临床试验结果公布）前后，短期期权的隐含波动率可能会显著高于长期期权。

B 选项错误。

- 波动率曲线的陡峭程度不一定随着到期时间的增加而增加。在重大事件前后，短期期权的波动率曲线可能会更陡峭。

C 选项正确。

- 在重大事件（如临床试验结果公布）前后，短期期权的波动率曲线会形成陡峭的微笑，因为市场预期价格会有大幅波动。而长期期权的波动率曲线则相对平坦，因为事件的影响会随时间减弱。

D 选项错误。

- 在重大事件前后，波动率曲线通常呈现微笑形状而不是 frown 形状，且短期期权的隐含波动率通常高于长期期权。

考察: 重大事件前后短期期权波动率曲线陡峭而长期期权平坦

20. A foreign exchange (FX) options analyst at a dealer bank is examining historical daily changes in several FX rates. The analyst compiles these daily changes into distributions that are used to determine the prices of FX options, which shape the options' implied volatility curves. The analyst also compares the typical implied distribution for FX options with a lognormal distribution. Which of the following observations regarding the distribution of FX rate changes is correct for the analyst to make?

- A. The excess kurtosis in FX rate distribution leads to higher prices for deep out-of-the-money put options than the prices derived using the lognormal distribution
- B. The lognormal distribution reflecting daily FX rate changes causes the implied volatility smile exhibited by foreign exchange options
- C. The negative skewness of FX rate distributions leads to higher implied volatilities for deep out-of-the-money put options than for deep out-of-the-money call options
- D. Distributions of historical daily FX rate changes generally have fewer observations close to the mean and more observations in the tails than a lognormal distribution

答案:A

解析:

A 选项正确。

- 汇率分布的超额峰度（excess kurtosis）表明极端价格变动的概率高于对数正态分布。这导致深度价外看跌期权的价格高于使用对数正态分布计算的价格。

B 选项错误。

- 对数正态分布本身不会导致波动率微笑。波动率微笑是由于实际分布与对数正态分布的差异（如超额峰度）导致的。

C 选项错误。

- 货币期权的波动率微笑通常是对称的，深度价外看跌期权的隐含波动率不一定高于深度价外看涨期权。

D 选项错误。

- 汇率变动的历史分布通常比对数正态分布有更多的均值附近观测值和更少的尾部观测值，而不是相反。

考察: 汇率分布的超额峰度导致深度价外看跌期权价格更高

21. A financial analyst at a wealth management firm is using the Merton model to value the debt and the equity of NET, a digital media company. NET’ s capital structure consists of senior debt, subordinated debt, and equity. The analyst notices that the company currently faces financial difficulties, resulting in its total asset value being almost equal to the value of its senior debt. A manager at the wealth management firm informs the analyst that NET’ s asset volatility is likely to increase and that there is an equal probability of NET being acquired by a larger competitor or filing for bankruptcy. Assuming the manager’ s information is correct, what is the most likely valuation consequence for NET?

- A. The values of equity and subordinated debt will increase, but the value of senior debt will decrease
- B. The values of equity and senior debt will increase, but the value of subordinated debt will decrease
- C. The values of equity and senior debt will decrease, but the value of subordinated debt will increase
- D. The value of equity will increase, but the values of senior debt and subordinated debt will decrease

答案:A

解析:

A 选项正确。

- 资产波动率增加和可能的收购或破产会导致权益和次级债务的价值增加，因为这两种情况都可能带来更高的回报。而高级债务的价值会下降，因为波动率增加增加了违约风险。

B 选项错误。

- 高级债务的价值不会增加，因为资产波动率增加会增加违约风险，降低高级债务的价值。

C 选项错误。

- 权益的价值不会下降，因为可能的收购或破产都可能带来更高的回报。

D 选项错误。

- 次级债务的价值不会下降，因为可能的收购或破产都可能带来更高的回报。

考察: 资产波动率增加时权益和次级债务价值上升而高级债务价值下降